PES UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Software Engineering Lab (PES_SE_B_2026B)

Lab 6: Code and Branch Coverage Analysis

Student Name:	Deeksha G
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Assigned Program:	student_grades.py (Repo 16)
GitHub Repository:	https://github.com/DEEKSHAGURURAJA/L6_L7_16_student_grades

1. Objective

To measure, analyze, and achieve target test coverage on the assigned Python module (student_grades.py) using Python's coverage and pytest tools. The primary goals are to understand the practical difference between statement (line) coverage and branch (decision) coverage, achieve at least 90% in both, and commit the solutions according to the lab rubric.

2. Program Under Test (student_grades.py)

The application contains three core functions with multiple conditional branches: • **letter_grade(marks):** Converts marks (0 to 100) into letter grades (S, A, B, C, D, E, F). Raises ValueError for out-of-range marks (< 0 or > 100).
• **final_marks(internal, external, has_project):** Sums internal (max 40) and external (max 60) marks. Adds a 5-mark project bonus if applicable, capped at 100.
• **exam_eligibility(attendance, fees_paid, has_medical):** Evaluates student eligibility ('eligible', 'detained', or 'blocked') based on fee status, attendance percentage, and medical condonation.

3. Summary of Deliverables

• **Code Coverage Suite:** CodeCoverageTest_student_grades.py — achieves 100% statement coverage.
• **Branch Coverage Suite:** BranchCoverageTest_student_grades.py — achieves 100% branch coverage (26/26 branches).
• **Version Control:** Two distinct commits created and pushed upstream to GitHub.

4. Part 1: Code Coverage (Statement Coverage)

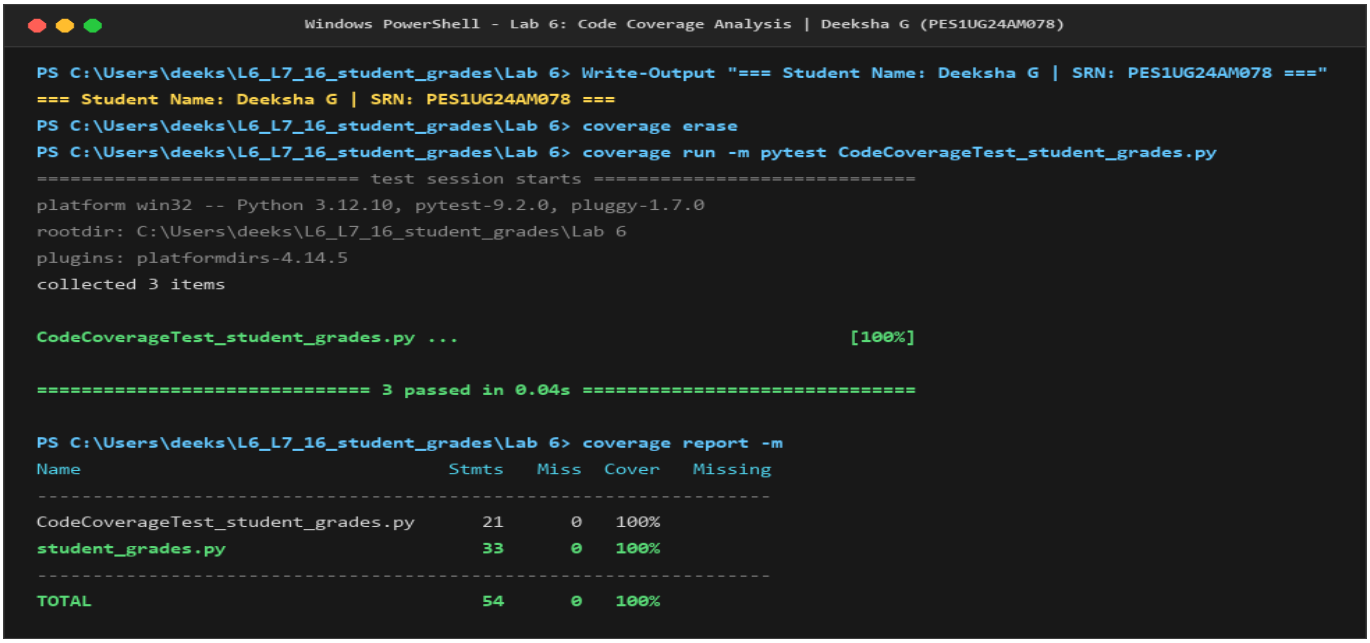
Test File: CodeCoverageTest_student_grades.py
Focus: Execute 100% of executable statements in student_grades.py.

Test Design:
• test_letter_grade_all_statements: Tests negative marks (-5) for ValueError, and representative values (95, 85, 75, 65, 55, 45, 35) to execute all grade return statements.
• test_final_marks_all_statements: Tests invalid internal marks (45 > 40) for ValueError, and (40, 60, True) to exercise bonus addition and the 100-cap statement.
• test_exam_eligibility_all_statements: Tests all return branches: fees unpaid ('blocked'), high attendance ('eligible'), medical condonation ('eligible'), and low attendance ('detained').

Commands Run:

```
coverage erase
coverage run -m pytest CodeCoverageTest_student_grades.py
coverage report -m
```

Terminal Screenshot (Code Coverage - 100% Statements Covered):



5. Part 2: Branch Coverage Analysis

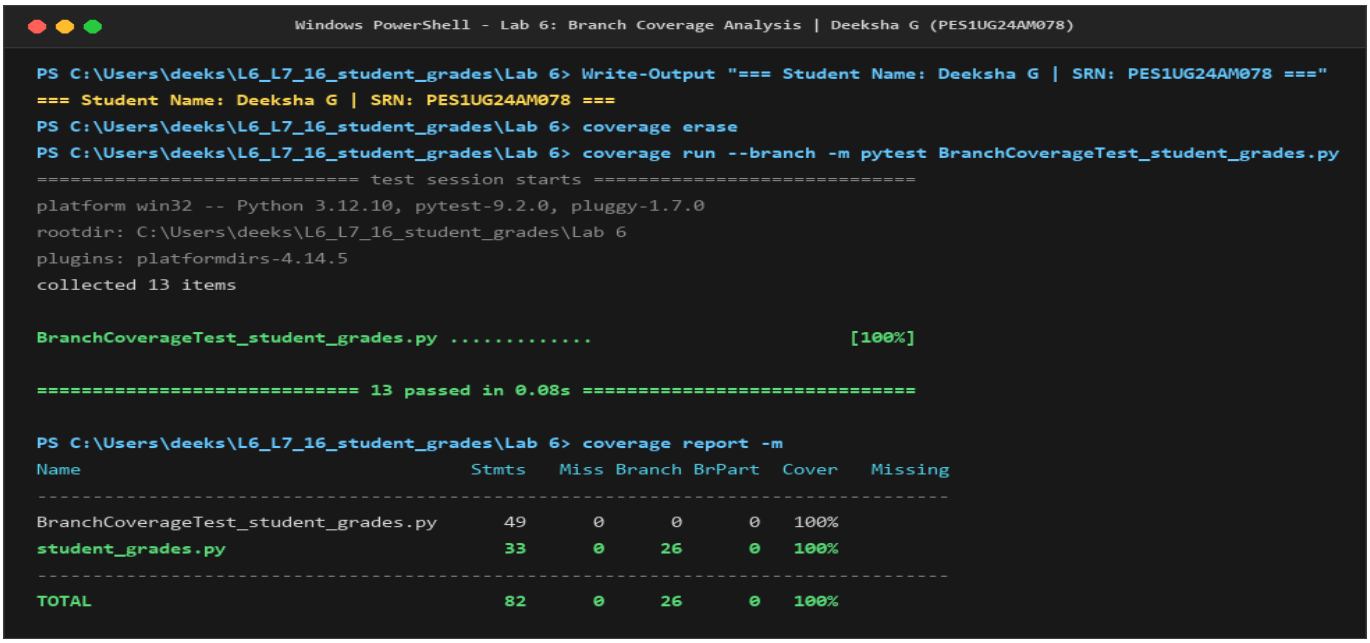
Test File: BranchCoverageTest_student_grades.py
Why Line Coverage Was Not Enough: Line coverage can create a false sense of security. In Part 1, 100% statements were executed, yet several logical branches were unexercised (e.g. has_project=False, total <= 100 without cap, marks > 100, and attendance between 65-75 without medical condonation). Branch coverage requires every condition to evaluate to both **True** and **False**.

Targeted Test Functions (13 tests covering all 26 branches):
• test_letter_grade_invalid_negative & test_letter_grade_invalid_exceeds_100: Evaluates both True branches of the OR condition.
• test_letter_grade_boundaries_and_ranges: Evaluates boundary values for each grade range (S: 100, 90; A: 89, 80; B: 79, 70; C: 69, 60; D: 59, 50; E: 49, 40; F: 39, 0).
• test_final_marks_*: Evaluates invalid internal (>40), invalid external (>60), project=False, project=True (total<=100), and project=True (total>100).
• test_exam_eligibility_*: Evaluates unpaid fees, high attendance (>=75), condonation with medical (True+True), condonation without medical (True+False), and low attendance (<65).

Commands Run:

```
coverage erase
coverage run --branch -m pytest BranchCoverageTest_student_grades.py
coverage report -m
```

Terminal Screenshot (Branch Coverage - 100% Branches Covered):



6. Detailed Branch Coverage Breakdown

Breakdown of all 26 decision branches in student_grades.py:

Function	Decision / Condition	Branches Evaluated	Status
letter_grade	marks < 0 or marks > 100	<0 (True), >100 (True), Valid range (False)	100%
	marks >= 90, 80, 70, 60, 50, 40	Each tier tested for True and False boundaries	100%
	Else (fall-through)	Grade F (tested at 39 and 0)	100%
final_marks	internal > 40 or external > 60	internal > 40, external > 60, valid marks	100%
	has_project	True (+5 bonus), False (no bonus)	100%
	total > 100 (with project)	True (capped at 100), False (uncapped)	100%
exam_eligibility	not fees_paid	True ('blocked'), False (proceeds)	100%
	attendance >= 75	True ('eligible'), False (< 75)	100%
	attendance >= 65 and has_medical	True+True ('eligible'), True+False ('detained'), False ('detained')	100%

7. Version Control & Git Commits

8. Conclusion